

Omar Sherif Elghamry

Backend & AI Engineer

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SUMMARY

Backend Software Engineer focused on Node.js/NestJS and Python/FastAPI, with hands-on production experience building and integrating real systems used by clients. Skilled in scalable REST APIs, payment integrations, event-driven architecture with Apache Kafka and RabbitMQ, real-time features with Socket.IO, and AI integration using LangChain, LangGraph, RAG, and AWS Bedrock. Comfortable across the full delivery lifecycle: PostgreSQL and Redis, Docker and Kubernetes (Amazon EKS), and CI/CD with GitHub Actions. Applies clean architecture, SOLID, and design patterns consistently. Currently deepening distributed systems, system design, observability, and performance. Strong computer science foundation with GPA 3.8.

EXPERIENCE

Software Engineer — Techkhana

2026 – Present | Full-time

- Backend engineer on **Barns**, a Saudi-market super-app, working across multiple production services — payments, e-commerce, event-driven data, access control, media, and AI — used by real clients.
- **Payments**: refactored the payment architecture around a provider abstraction (PaymentProviderFactory / strategy) so multiple providers sit behind one interface; integrated Tap Payments alongside PayFort, implementing charge, save-card, and authorize flows with card tokenization and platform-specific configuration (X-Platform) for mobile and web.
- **E-commerce checkout**: designed a checkout that stores a snapshot of catalog data (price, quantity, product metadata) at checkout time, decoupling orders from mutable catalog data and enforcing a clean Checkout → Order → Payment separation.
- **Event-driven data**: worked with Apache Kafka and Kafka Connect (CDC) for change-data-capture synchronization; implemented backend-to-backend communication between services, including the app-to-admin support-request flow and menu-module backend architecture.
- **Caching**: used Redis for cache-aside caching, TTL management, and cache invalidation to reduce load and latency.
- **Auth & access control**: built admin role-based access control (roles, permissions, guards) with a layered request pipeline of authentication → authorization → throttling.
- **Image processing**: built a centralized ImageProcessingService — magic-byte validation, EXIF stripping, WebP conversion, SHA-256 deduplication, and immutable caching over S3; validates size, type, dimensions, and count, and rejects unsupported formats (video, HEIC).
- **AI feature ("Fahem")**: integrated AWS Rekognition for image analysis with feature flags, per-user daily usage limits, image validation, product matching, and asynchronous usage tracking.
- **Infrastructure & delivery**: ran services on Kubernetes (Amazon EKS) with Docker and GitHub Actions CI/CD; handled production debugging, deployment, and launch support.

Software Engineer Intern — Techkhana

June 2025 – November 2025 | Hybrid | 360 hours

- Primary backend engineer on an Agentic AI service — Python and Node.js, AWS Bedrock, multi-agent orchestration and RAG pipelines, with PostgreSQL persistence; resolved performance bottlenecks and model inconsistencies. Delivered an internal session on Introduction to Deep Learning for the engineering team.
- Built RESTful APIs and business-logic modules for a Construction Management Platform (Node.js, PostgreSQL); implemented features and fixed client-critical bugs in NestJS services with Testcontainers end-to-end tests.
- Built and maintained CI/CD pipelines with GitHub Actions; containerized services with Docker and Docker Compose; worked in an Agile team using Jira and GitHub pull requests and code reviews.

GRADUATION PROJECT

The Smart Fashion Platform (StyleHub) — Misr University for Science and Technology (MUST)

2025 – 2026 | Grade: A*

- Microservices platform combining e-commerce, social media, and AI-driven recommendations; the hybrid Neural Collaborative Filtering + content-based engine achieved AUC-ROC 0.93 and 92.87% accuracy. Services communicate via RabbitMQ and deploy on Heroku.
- Sole owner of the Social Media Service backend (NestJS, TypeScript, PostgreSQL, TypeORM) — post management with media uploads to Cloudinary, follow/unfollow with atomic counters, likes and comments with transaction-safe sync, and a fan-out-on-write feed. 40+ REST endpoints with RBAC and rate limiting; 12 test suites / 161 tests passing.
- Designed the inter-service messaging layer on a RabbitMQ durable topic exchange (auto-reconnect, offline buffering, event-to-routing-key mapping) and a Socket.IO gateway with JWT auth, per-user rooms, real-time notifications, and typing indicators. Managed 13 TypeORM migrations for zero-downtime schema evolution.
- Built the Python/FastAPI AI service (LangChain, LangGraph, RAG, multi-agent over OpenAI and AWS Bedrock) and the Flutter mobile app for the social service.

EDUCATION

TECHNICAL SKILLS

Backend: Node.js, NestJS, TypeScript, Python, FastAPI, Express

Databases & Caching: PostgreSQL, TypeORM, Redis, SQL

Messaging & Events: Apache Kafka, Kafka Connect (CDC), RabbitMQ, Event-Driven Architecture, Topic Exchange

Real-time: Socket.IO, WebSocket, Per-User Rooms, Typing Indicators

Payments: Tap Payments, PayFort, Provider Abstraction, Card Tokenization

APIs & Auth: REST API Design, JWT, Role-Based Access Control, Rate Limiting

AI & ML: LangChain, LangGraph, RAG, OpenAI API, AWS Bedrock, AWS Rekognition, Agentic AI, Deep Learning

DevOps & Cloud: Docker, Docker Compose, Kubernetes (Amazon EKS), GitHub Actions, CI/CD, AWS (S3, EKS, Bedrock, Rekognition), Heroku

Testing: Jest, Testcontainers, Unit & End-to-End Testing, Postman

Principles: Clean Architecture, SOLID, Design Patterns, Microservices, OOP, Data Structures & Algorithms

Tools: Git, GitHub, Jira, Slack

COURSES AND TRAINING

- DeepLearning.AI — Machine Learning & Deep Learning Specialization: supervised/unsupervised learning, neural networks, deep learning fundamentals.
- Udemy — Complete Agentic AI Bootcamp: LangChain, LangGraph, multi-agent orchestration, and RAG pipelines.
- Udemy — Amazon Bedrock: The Complete Guide to AWS Generative AI: foundation models and enterprise AI integration.
- Udemy — NestJS: The Complete Developer Guide: modular architecture, REST APIs, authentication, and backend testing.
- Udemy — Mastering Data Structures and Algorithms in C++: algorithmic thinking and problem-solving.

LANGUAGES

Arabic: Native | English: Excellent